Collaborative Filtering for MeSH Keyword Recommendation on Clinical Trial Records

UE24CS352A – Machine Learning · Mini-Project Write-up · Team: Ramneek Gautam (PES1UG24AM220), Rufus Thomas (PES1UG24AM232) · Problem No. [__]

1. Problem Statement

Researchers searching ClinicalTrials.gov for similar studies usually give a short query that omits many relevant medical terms, and keyword-based retrieval (BM25, LSH) can only match what the query contains. We build a **keyword recommender**: given the medical (MeSH) keywords found in a query, it predicts further keywords the user probably forgot, which can then be added to the query for retrieval. The project re-implements the approach of Zhou (CS229, 2020), which treats trials as "users", MeSH terms as "items" and TF-IDF as the "rating".

2. Dataset

- **Source:** ClinicalTrials.gov API v2 (National Library of Medicine). **50,000 trial records** were downloaded; for each we used the brief and official title, brief summary, detailed description, conditions and eligibility criteria.
- **Vocabulary:** NLM Medical Subject Headings (MeSH) 2026 descriptor file, with all entry-term synonyms mapped to one concept.
- **Utility matrix:** 50,000 documents × **1,000 MeSH terms** (the most frequent terms; no frequency cutoff). The matrix is **very sparse**: each trial contains only a small fraction of the 1,000 terms.
- **Split:** documents (columns) are split randomly 80% / 10% / 10% into train, dev and test sets.

3. Approach

Keyword extraction. MeSH terms are loaded into a word-level **Trie**. Text is scanned left to right with longest-match-first, so "breast cancer" wins over "cancer", and synonyms such as "mammary carcinoma" map to the same concept.

TF-IDF utility matrix. $U_{ij} = \log_{10}(1 + \text{count of term i in doc j}) \cdot \log_{10}(N / df_i)$. Values are non-negative and zero exactly when the term is absent.

Model. The TF-IDF of term i in a document is estimated from the other terms in the same document: $\hat{R}_{ij} = \text{ReLU}(\Sigma_f W_{if} \Theta_{if} R_{fj} + b_i)$. W is a *fixed* neighbour-weight matrix, $W_{if} = \exp(\tau (1 - \cos(i,f))^k)$, computed from the cosine similarity of term rows, with $W_{ii} = 0$ so a term cannot predict itself. Θ and b are learned. ReLU keeps predictions non-negative like TF-IDF. The loss is $\frac{1}{2}\Sigma\Sigma(\hat{R} - R)^2 + \frac{1}{2}\lambda\|\Theta\|^2$, minimised with mini-batch gradient descent using the gradients derived in the paper.

Evaluation. For each test document, 50% of its non-zero terms are masked to zero; the model sees only the rest and ranks the masked-out candidates. **APTK** = average precision of the top K = 3 recommended terms (the fraction that were truly hidden terms). **APRK** is the same with K random terms. MSE and MSER (squared error divided by the number of relevant cells) measure reconstruction. We added two baselines not in the paper: **popularity** (always recommend the most frequent absent terms) and **item-item cosine kNN** (k = 50).

4. Implementation Overview

- **extract.py** – MeSH parser and Trie extractor. **pipeline.py** – data download, TF-IDF matrix, model, training, evaluation, baselines and hyperparameter search. **recommend.py / app.py** – a Streamlit demo: the user pastes a query, sees the MeSH keywords found, and gets the top-K recommended additional keywords.
- Pure **NumPy** implementation with hand-derived gradients (ReLU′(0) = 1 as in the paper). Mini-batch size 512, learning rate 0.1, **60 epochs**; the loss flattened by epoch ~55. τ, k and λ were selected by a random search on the dev set (30 configurations).

5. Results (test set)

Model	MSE	MSER	APTK (K=3)
Random selector (APRK)	–	–	0.0152
Popularity baseline	–	–	0.4476
Item-item kNN (k=50)	–	–	0.4162
CF + ReLU (ours)	**0.00960**	**0.00994**	**0.4776**

MSE/MSER are not reported for kNN and the simple baselines because their scores are not on the TF-IDF scale; compare them on APTK only.

6. Conclusions and Limitations

- The model recommends useful keywords: top-3 precision is **0.478**, about **31× random**, **+3.0 points over the popularity baseline** and **+6.1 points over item-item kNN**.
- Popularity is a strong baseline (0.448) because frequent terms are easy to predict, so the margin over it is modest. All results come from a single split with no confidence intervals, so we claim a modest, not a decisive, improvement. Since W is fixed, W⊙Θ is effectively one learned matrix; the model behaves like a regularised linear autoencoder with a zero diagonal.
- **Not directly comparable to the paper's APTK of 0.50:** the paper used 15,071 terms and did not report a popularity baseline, whereas we use the 1,000 most frequent terms; the two scores are in the same range. An earlier run using the registry's own MeSH labels *including ancestor terms* reached APTK 0.72, but that is inflated because

parent terms are almost certain once a child term is present, so the Trie-based result is the fair one.

- **Limitations:** matching is lower-cased (the paper's "Protein A" vs "A" distinction is lost) and some plain words match MeSH synonyms; only 1,000 terms, due to the $O(m^2)$ cost; generic terms are sometimes recommended; evaluation is on masked terms, not end-to-end retrieval. **Future work:** more terms, a stoplist for ambiguous synonyms, and measuring retrieval quality with the recommended keywords added.

Reference: X. Zhou, "Collaborative Filtering on Keywords Recommendation for Clinical Trial Records", Stanford CS229 project report, 2020. Data: ClinicalTrials.gov and NLM MeSH (2026).